

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		{Provide {nutrients/ named nutrient}/ nourishes} for {developing cells/ pollen/ tetrad}/ { provides waterproof/ outer coat} for pollen grains/ protects pollen grain from drying out	1			1		
		(ii)		(Pollen mother cell divides by) meiosis (1) Forming four (haploid) {cells/ nuclei } (1) NOT diploid Ignore reference to pollen		2		2		2
	(b)	(i)		{Controls/ directs } pollen tube. (1) {controls synthesis of/ codes for} {enzymes/ hydrolytic enzymes}. (1) Reject makes/ contains/ secretes	2			2		
		(ii)		0.2 = 3 marks If incorrect award 2 marks for 0.17 [mm minute ⁻¹] 0.166666666666.... 1/6 If incorrect award 1 mark for $\frac{20 \times 10}{20 \times 60}$ (conversion)		3		3	3	
		(iii)		Both A and B correct (1) Candidate must have an indication of both polar nuclei		1		1		1
		(iv)		Any three × (1) from: A. <u>megaspore</u> (mother cell) undergoes meiosis.(1) B. forming four haploid {cells/ nuclei} (1) C. One undergoes three mitotic divisions (1) D. One of these (cells/ nuclei) becomes the {gamete/ oosphere/ egg cell/ ovum} (1)	3			3		1

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		(v)		Any three × (1) from: A. 1 {male gamete/ nucleus} fuses with {gamete/ oosphere/ egg cell/ ovum} (1) NOT oocyte B. to produce a { <u>diploid/ 2n</u> } zygote (1) C. Second male gamete fuses with (two) polar nuclei (1) D. to produce { <u>triploid/ 3n</u> } endosperm (nucleus)(1)		3		3		
				Question 1 total	6	9	0	15	3	4